

Enterprise Database Infrastructure Administration Lab

Design, Deployment, Security Hardening, Backup, Replication and
Monitoring of PostgreSQL Infrastructure on Linux

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AUTOMATION & OPERATIONS ENGINEERING

I. Objective

The goal of this deliverable is to automate repetitive operational tasks and establish standardized operational procedures.

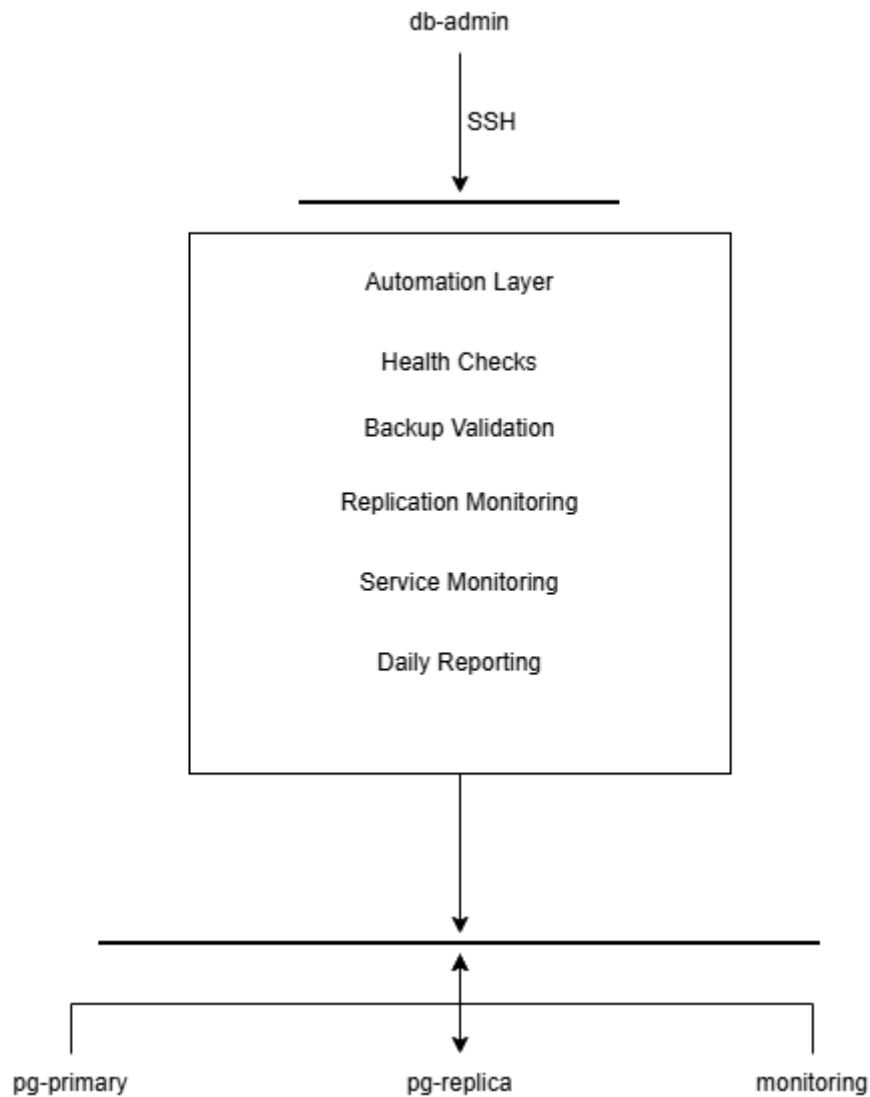
In a professional environment, administrators do not manually execute every task. They automate :

- Health checks
- Service validation
- Backup verification
- Replication monitoring
- Storage monitoring
- Incident detection
- Reporting

At the end of this phase :

- Operational scripts created
- Health checks automated
- Backup validation automated
- Replication checks automated
- Daily reporting implemented
- Operational runbooks documented
- Ansible foundation introduced

II. Automation Architecture



III. Create Operations Directory

On db-admin :

```
mkdir -p ~/operations
```

Create structure :

```
mkdir -p ~/operations/scripts
```

```
mkdir -p ~/operations/logs
```

```
mkdir -p ~/operations/reports
```

```
mkdir -p ~/operations/runbooks
```

Verify :

```
tree ~/operations
```

IV. PostgreSQL Service Health Check

Create :

```
nano ~/operations/scripts/check_postgresql.sh
```

Content :

```
#!/bin/bash
HOST="pg-primary"
ssh david@$HOST \
"systemctl is-active postgresql"
```

Permissions :

```
chmod +x \
~/operations/scripts/check_postgresql.sh
```

Test :

```
./check_postgresql.sh
root@db-admin:/operations/scripts# ./check_postgresql.sh
active
root@db-admin:/operations/scripts#
```

V. Multi-Server Health Check

Create :

```
nano ~/operations/scripts/check_all_servers.sh
```

Content :

```
#!/bin/bash
SERVERS=(
pg-primary
pg-replica
monitoring
)
for SERVER in "${SERVERS[@]}"
do
echo "Checking $SERVER"
ssh david@$SERVER \
"hostname && uptime"
echo "-----"
```

done

Test :

`./check_all_servers.sh`

```
root@db-admin:/operations/scripts# ./check_all_servers.sh
Checking pg-primary
pg-primary
19:29:25 up 5 min, 3 users, load average: 0,05, 0,45, 0,29
-----
Checking pg-replica
pg-replica
19:29:29 up 0 min, 1 user, load average: 1,34, 0,41, 0,14
-----
Checking monitoring
monitoring
19:29:30 up 4 min, 3 users, load average: 1,02, 0,95, 0,45
-----
root@db-admin:/operations/scripts#
```

VI. PostgreSQL Connectivity Check

Create :

`nano ~/operations/scripts/check_database.sh`

Content :

```
#!/bin/bash

PGPASSWORD="NewStrongPassword2026!"

psql \
-h pg-primary \
-U app_user \
-d company_prod \
-c "SELECT NOW();"
```

Test :

`./check_database.sh`

```
root@db-admin:/operations/scripts# psql --version
psql (PostgreSQL) 16.14 (Ubuntu 16.14-0ubuntu0.24.04.1)
root@db-admin:/operations/scripts# ./check_database.sh
Password for user app_user:
now
-----
2026-06-19 19:35:44.591102+00
(1 row)
root@db-admin:/operations/scripts#
```

VII. Replication Status Automation

One of the most valuable DBA automation tasks.

Create :

`nano ~/operations/scripts/check_replication.sh`

Content :

```
#!/bin/bash

ssh david@pg-primary \

"sudo -u postgres psql -At \
```

```
-c \"SELECT state
FROM pg_stat_replication;\""
```

Test :

```
./check_replication.sh
```

VIII. Replication Lag Check

Create :

```
nano ~/operations/scripts/check_replication_lag.sh
```

Content :

```
#!/bin/bash
ssh david@pg-primary \
"sudo -u postgres psql -c \"
SELECT
client_addr,
write_lag,
flush_lag,
replay_lag
FROM pg_stat_replication;
\""
```

IX. Backup Verification Automation

Create :

```
nano ~/operations/scripts/check_backups.sh
```

Content :

```
#!/bin/bash
find \
/backups/postgresql/logical \
-type f \
-mtime -1
```

X. Storage Monitoring Script

Create :

```
nano ~/operations/scripts/check_disk_usage.sh
```

Content :

```
#!/bin/bash
ssh david@pg-primary \
"df -h /"
ssh david@pg-replica \
"df -h /"
ssh david@monitoring \
"df -h /"

root@db-admin:/operations/scripts# ./check_disk_usage.sh
=====PG-PRIMARY=====
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 19G   8,5G   9,2G  48% /
=====PG-REPLICA=====
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 19G   7,0G   11G  40% /
=====MONITORING=====
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 19G   8,9G   8,7G  51% /
root@db-admin:/operations/scripts#
```

XI. Memory Monitoring Script

Create :

```
nano ~/operations/scripts/check_memory.sh
```

Content :

```
#!/bin/bash
for SERVER in \
pg-primary \
pg-replica \
monitoring
do
echo "=== $SERVER ==="
ssh david@$SERVER \
"free -h"
done
```

```

root@db-admin:/operations/scripts# ./check_memory.sh
=== pg-primary ===
Mem:      total      used      free      shared  buff/cache  available
Swap:    3,8Gi      526Mi      2,8Gi      15Mi      709Mi      3,3Gi
=== pg-replica ===
Mem:      total      used      free      shared  buff/cache  available
Swap:    3,8Gi      550Mi      3,1Gi      13Mi      394Mi      3,2Gi
=== monitoring ===
Mem:      total      used      free      shared  buff/cache  available
Swap:    3,8Gi      658Mi      2,6Gi      1,3Mi      759Mi      3,1Gi
root@db-admin:/operations/scripts#

```

XII. Database Size Monitoring

Create :

```
nano ~/operations/scripts/check_db_size.sh
```

Content :

```

#!/bin/bash
ssh david@pg-primary \
"sudo -u postgres psql \
-c \"
SELECT
pg_size_pretty(
pg_database_size('company_prod')
);
\"\"

```

```

root@db-admin:/operations/scripts# ./check_db_size.sh
pg_size_pretty
-----
7815 kB
(1 row)
root@db-admin:/operations/scripts#

```

XIII. Daily Operations Report

Create :

```
nano ~/operations/scripts/daily_report.sh
```

Content :

```

#!/bin/bash
REPORT=\
~/operations/reports/report_$(date +%F).txt
echo "Daily Operations Report" > $REPORT
date >> $REPORT

```

```
echo "" >> $REPORT
echo "Database Status" >> $REPORT
~/operations/scripts/check_database.sh \
>> $REPORT
echo "" >> $REPORT
echo "Replication Status" >> $REPORT
~/operations/scripts/check_replication.sh \
>> $REPORT
```

Execute :

```
./daily_report.sh
```

Verify :

cat \

```
~/operations/reports/report_*.txt
```

```
root@db-admin:/operations/scripts# ./daily_report.sh
Password for user app_user:
root@db-admin:/operations/scripts# ll ../reports/
total 12
drwxr-xr-x 2 root root 4096 juin  21 10:03 ./
drwxr-xr-x 6 root root 4096 juin  19 18:40 ../
-rw-r--r-- 1 root root  196 juin  21 10:03 report_2026-06-21.txt
root@db-admin:/operations/scripts# cat ../reports/report_2026-06-21.txt
Daily Operations Report
dim. 21 juin 2026 10:03:36 UTC

Database Status
              now
-----
2026-06-21 10:03:40.808406+00
(1 row)

Replication Status
root@db-admin:/operations/scripts#
```

XIV. Centralized Logging

Modify scripts to log execution.

Example :

```
echo \
"$(date): Check completed" \
>> ~/operations/logs/operations.log
```

XV. Scheduled Operations

Configure cron.

```
crontab -e
```

Database Check :

```
*/15 * * * * \ /operations/scripts/check_database.sh
```

Replication Check :

```
*/15 * * * * \ /operations/scripts/check_replication.sh
```

Daily Report :

```
0 6 * * * \ /operations/scripts/daily_report.sh
```

Backup Validation :

```
0 7 * * * \ /operations/scripts/check_backups.sh
```

XVI. Operational Runbooks

Create :

```
mkdir -p \
~/operations/runbooks
```

1. Runbook 1

File : postgresql_service_failure.md

Content :

```
# PostgreSQL Service Failure
## Symptoms
Cannot connect
## Investigation
systemctl status postgresql
journalctl -u postgresql
## Resolution
Restart service
Verify connectivity
```

2. Runbook 2

File : replication_failure.md

Content :

```
# Replication Failure
## Symptoms
Replication stopped
```

```
## Investigation
SELECT * FROM pg_stat_replication;
## Resolution
Restart replica
Rebuild if necessary
```

XVII. Introduction to Ansible

At this stage, your environment is ready for automation using : Ansible.

Install on db-admin :

```
sudo apt install ansible -y
```

Verify :

```
ansible --version
```

Inventory :

```
mkdir -p ~/ansible
nano ~/ansible/inventory.ini
```

Content :

```
[database_group]
pg-primary ansible_host=192.168.10.20 ansible_user=david
pg-replica ansible_host=192.168.10.21 ansible_user=david
[monitoring_group]
monitoring ansible_host=192.168.10.60 ansible_user=david
```

Test connectivity :

```
ansible all \
-i inventory.ini \
-m ping
```

```
root@db-admin:/ansible# ansible all -i inventory.ini -m ping
pg-primary | SUCCESS ==> {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
monitoring | SUCCESS ==> {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
pg-replica | SUCCESS ==> {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@db-admin:/ansible# nano inventory.ini
root@db-admin:/ansible#
```

XVIII. First Ansible Playbook

Create :

```
nano \
~/ansible/update_servers.yml
```

Content :

```
---
- hosts: all
  become: yes
  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes
    - name: Upgrade packages
      apt:
        upgrade: dist
```

Execute :

```
ansible-playbook \
-i inventory.ini \
update_servers.yml --ask-become-pass
```

```
root@db-admin:/ansible# ansible-playbook -i inventory.ini update_servers.yml --ask-become-pass
BECOME password:
PLAY [all] *****
TASK [Gathering Facts] *****
ok: [pg-primary]
ok: [pg-replica]
ok: [monitoring]
TASK [Update apt cache] *****
changed: [pg-primary]
changed: [pg-replica]
changed: [monitoring]
TASK [Upgrade packages] *****
ok: [pg-replica]
changed: [pg-primary]
changed: [monitoring]
PLAY RECAP *****
monitoring      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
pg-primary      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
pg-replica      : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
root@db-admin:/ansible#
```

XIX. Operations Dashboard Documentation

Document :

- **Daily Tasks**
 - Check backups
 - Check replication
 - Check PostgreSQL
 - Review Grafana
- **Weekly Tasks**
 - Restore test
 - Capacity review

- Log review
- **Monthly Tasks**
 - Password rotation
 - Disaster recovery test
 - Patch management